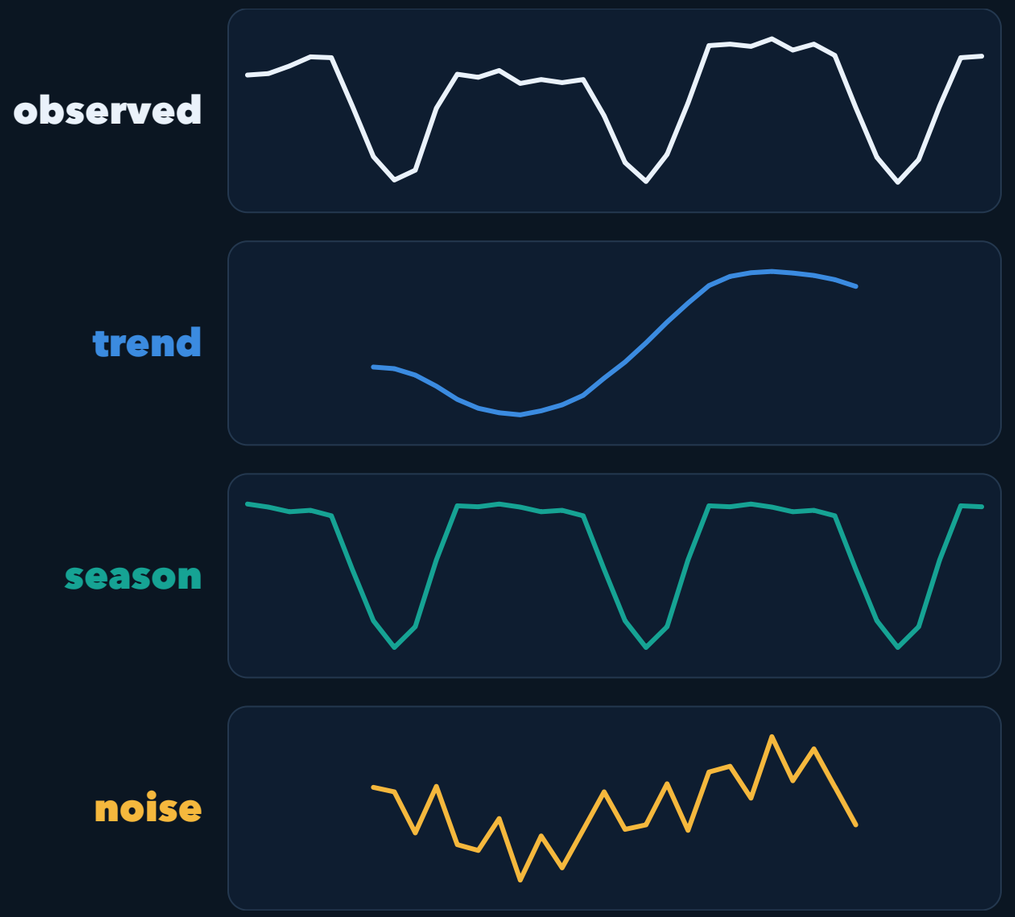
WEEK 2 · DAY 1

Take It Apart: Trend, Season and Noise

One line on a chart is really three stories stacked on top of each other. Today we pull them apart. 🔧



NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy

WEEK 1 IN 3 LINES

Quick warm-up before we open the engine



Order matters. Shuffle a time series and the story dies.



Averages can hide trouble. Look at the minimum when the question is “did we breach?”



A 7-day window smooths away the weekend rhythm.

Today's question: **that weekend rhythm and that rainy-season dip… can we measure them?**

TODAY'S MISSION · 2 HOURS

Three moves



Split it

trend + season + residual



Pick the rhythm

the period comes from the process



Read the leftovers

the residual is where trouble hides

⏱️ 10 min hook

🎤 30 min slides

👩🏾‍💻 60 min hands-on

🏁 20 min wrap-up

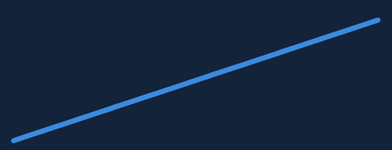
WHY IS YOUR TRIP TO WORK SO LONG?

Lagos traffic = three things added 🚦

📈

City growing

trend

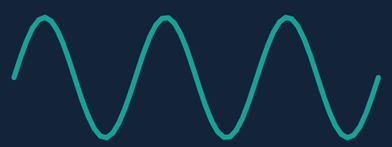


+

🔁

Rush hour, rainy season

seasonality



+

🛺

One keke breakdown

noise



=

🚗

Your trip to work

what you actually feel



Like jollof: you taste **one** pot, but it's rice + tomato base + that one surprise pepper. 🍛

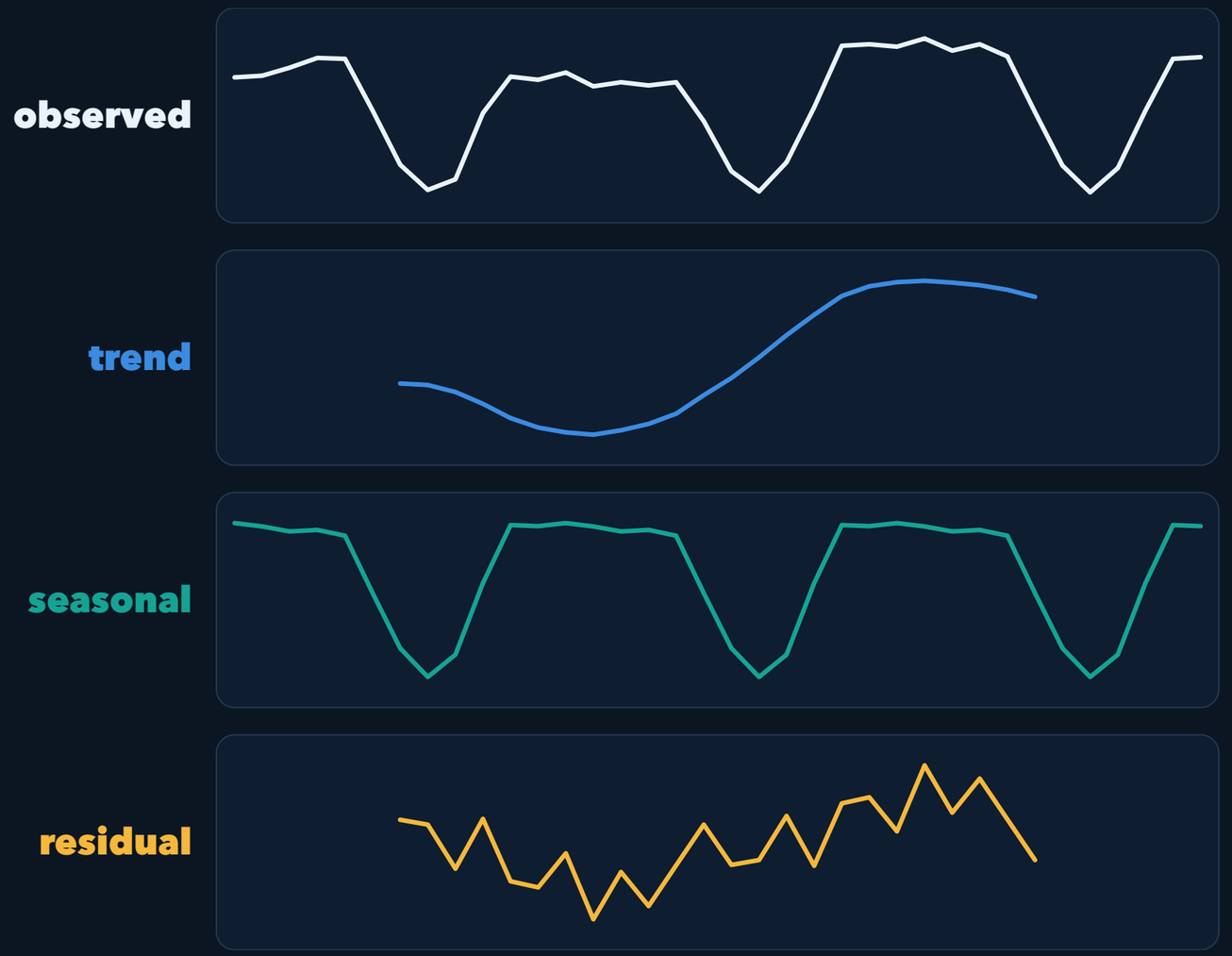
Decomposition = tasting each ingredient on its own.

THE ONE EQUATION

observed = trend + seasonal + residual

That's the **additive** model. Add the three bottom lines together and you get the top line back **exactly**.

🔁 **Seasonal** repeats every period
📈 **Trend** is what's left when the season is averaged away
🎲 **Residual** is whatever is left over



PART 1

One function does it all 🔧

statsmodels' `seasonal_decompose`: already installed in Colab.

THE RECIPE

Three lines, three parts

```
from statsmodels.tsa.seasonal import seasonal_decompose
parts = seasonal_decompose(kam["availability_pct"],
                           model="additive",    # or "multiplicative"
                           period=12)           # one rainy season = 12 months
parts.trend      # the slow direction
parts.seasonal   # the repeating pattern
parts.resid      # the leftovers
```

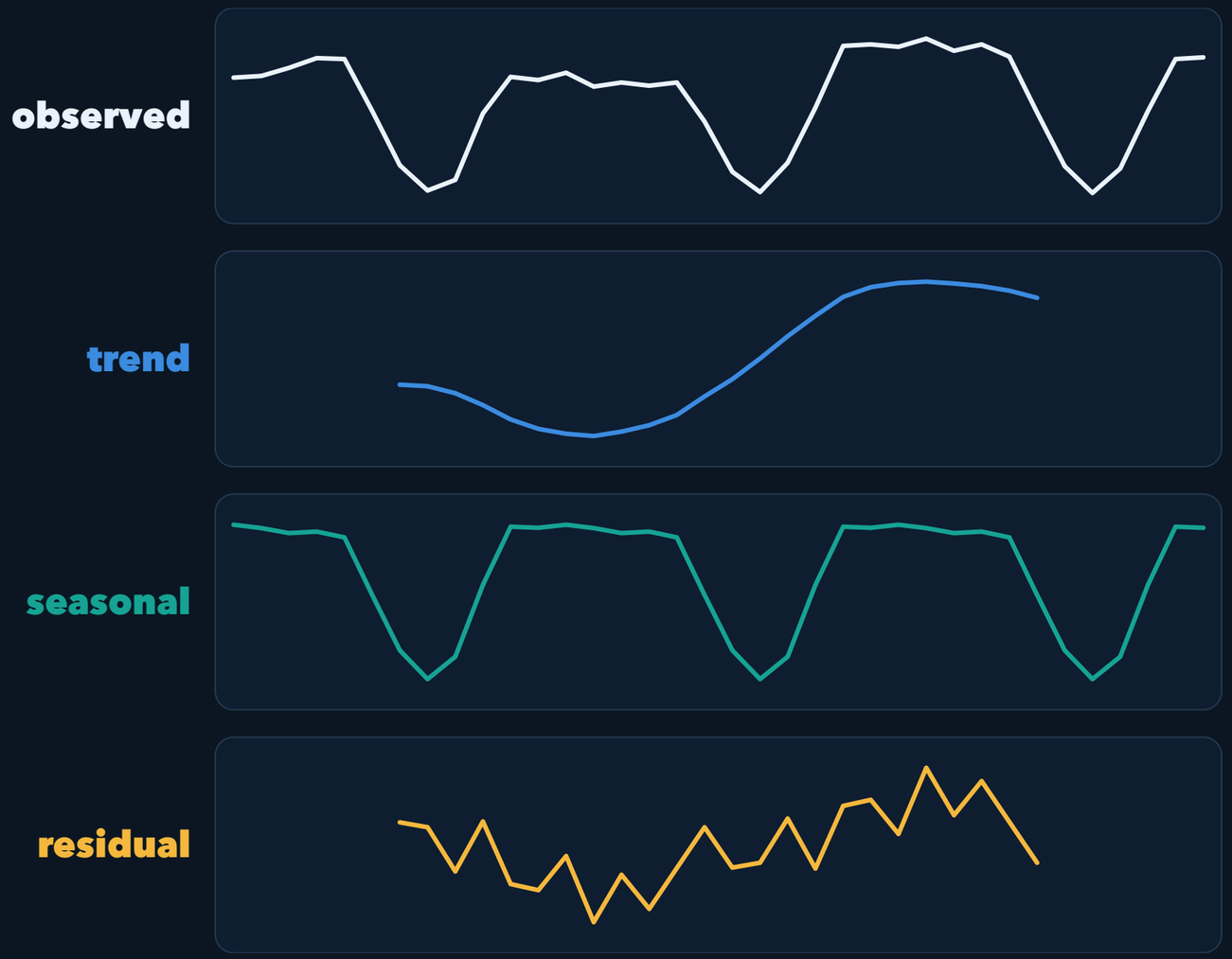
⚠ No gaps allowed

NaN inside the series → `ValueError`. Fill or trim first.

⚠ Two full cycles minimum

period=12 needs at least 24 rows, or it refuses to run.

KA-BAND AVAILABILITY, DECOMPOSED



🔁 Seasonal: worst month

Aug -1.1

points below the yearly average: rainy season

📈 Trend

99.24 → 99.40

slowly improving (ground-station upgrades)

🎲 Residual: biggest leftover

0.16

points. Nothing strange this time ✅

THE MOST COMMON MISTAKE

The period comes from the process, not a default



Bus power · hourly

24

the eclipse comes once a day, at 21:00 UTC



NOC tickets · daily

7

the working week repeats every 7 days



Ka-band · monthly

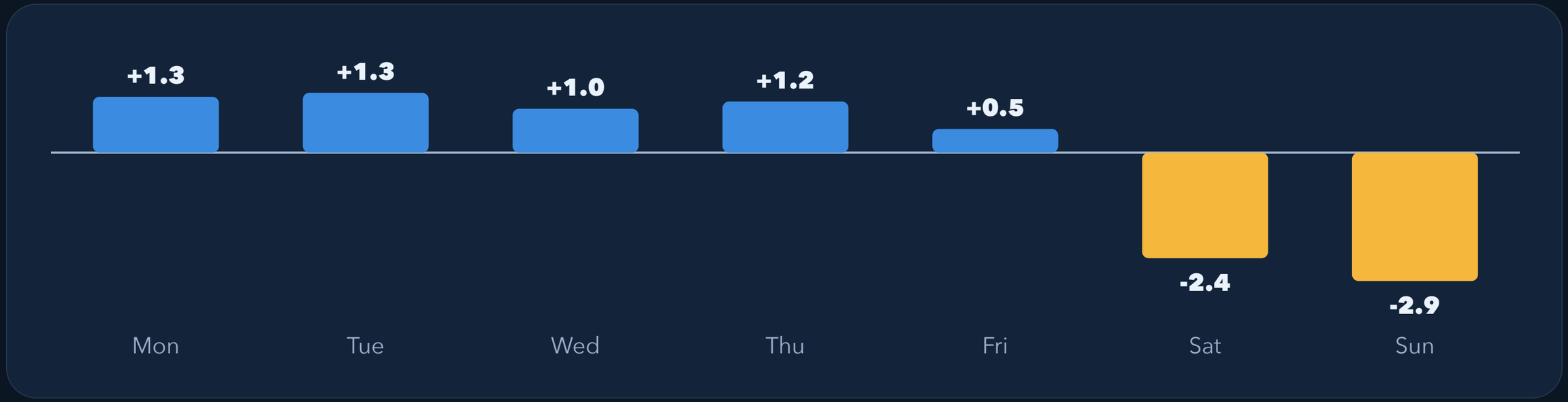
12

one rainy season every 12 months

Ask: **“how many rows until it repeats?”** That's your period.

NOC TICKETS · PERIOD = 7

Even faults rest on Sunday 😄 (now measured)



✅ **period=7**

noise left over: **1.23** tickets

❌ **period=12**

noise left: **2.27** tickets (weekend stuck in the noise)

PART 2

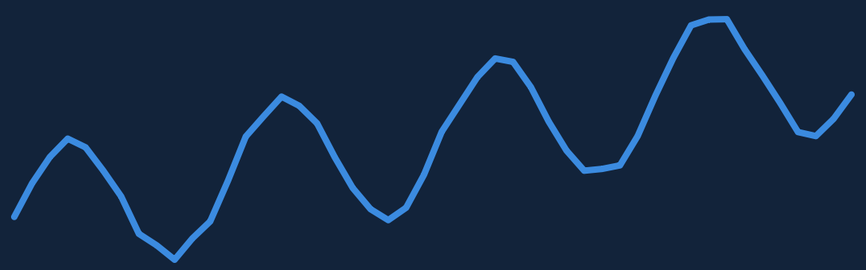
Add or multiply? ➕ ✖

Look at the swing. Does it stay the same size, or grow?

ADDITIVE_MULTIPLICATIVE_DEMO.CSV

Same trend, same season… different maths

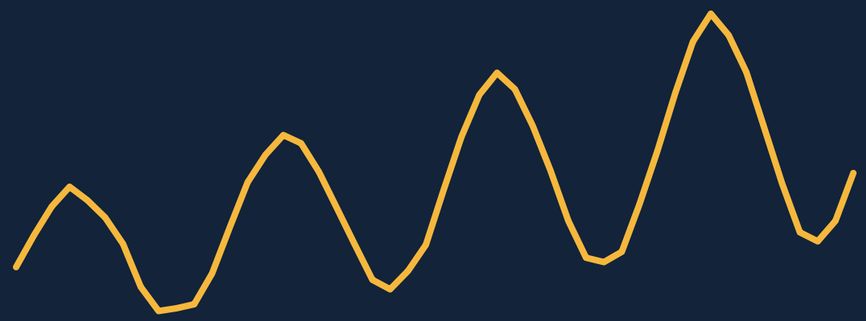
➕ Additive



year 1 year 4

Yearly swing: **0.29** → **0.28**. Stays the same size.
🚌 Like December fare going up **₦100** every year: a fixed amount.

✖ Multiplicative

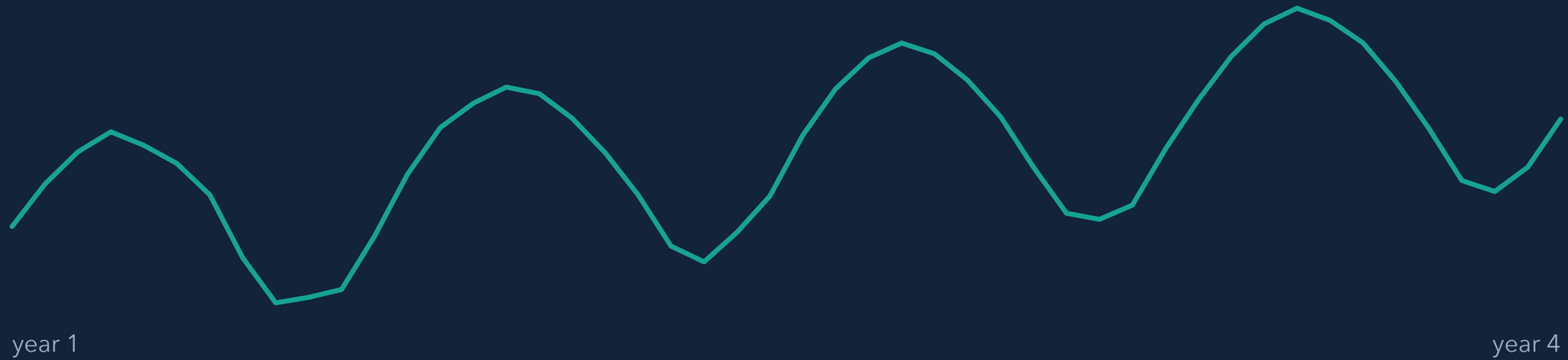


year 1 year 4

Yearly swing: **0.29** → **0.54**. Grows with the level.
🚌 Like December fare going up **20%**: 20% of a bigger fare is more naira.

THE LOG TRICK

Take the log, and multiply becomes add



swing before the log

0.29 → 0.54

swing after `np.log()`

0.77 → 0.83

PART 3

Read the leftovers 🔍

When trend and season are explained, what's left is the interesting part.

BUS POWER · SOLAR-ARRAY CURRENT · AUGUST · PERIOD = 24

The residual points straight at the fault 🎯



biggest leftover: 11.6 A

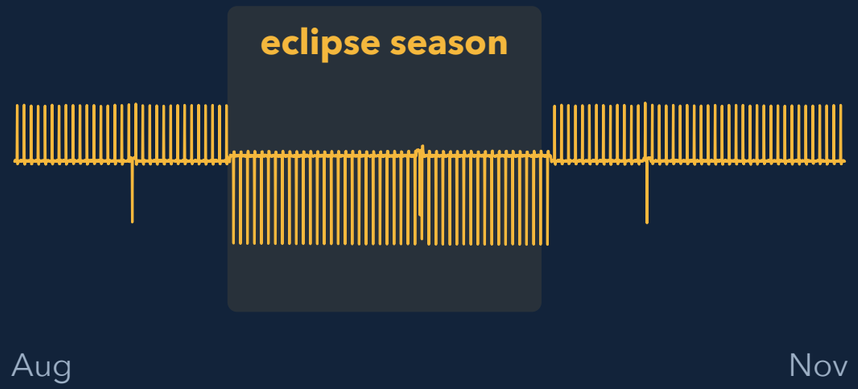
next biggest: only 0.59 A

NigComSat-1 was lost in 2008 to a solar-array failure. This is why we watch the array.

BUS POWER · ALL 4 MONTHS · PERIOD = 24

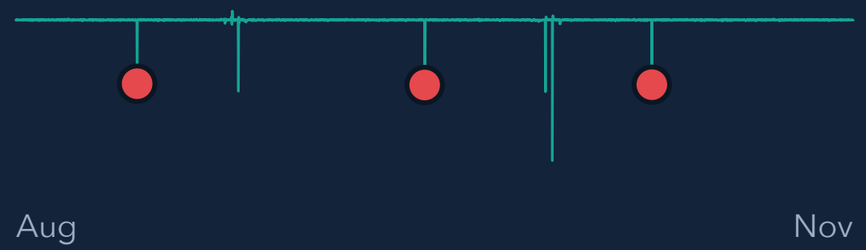
A season that comes and goes needs a smarter tool

😵 classic `seasonal_decompose`



Uses the **same** daily pattern for all 120 days, so the eclipse hours flood the residual. **46** of the top 46 leftovers are just eclipse.

🙌 `STL(robust=True)`



Lets the season change slowly. The injected array dropouts rank **#4–#7**; after them nothing is bigger than **1.75 A**.

WHAT IF ONE AUGUST WAS A DISASTER?

Don't let one bad month spoil every August



We faked **Aug 2024 = 94%**. Classic decomposition now says **every** August is -3.0.

`STL(robust=True)` keeps August at -1.2 and puts the disaster in the residual (-4.0). ✅

One bad experience at one buka shouldn't make you blacklist every buka in Lagos. 🍲

HOW MUCH HISTORY DO YOU NEED?

At least three full cycles

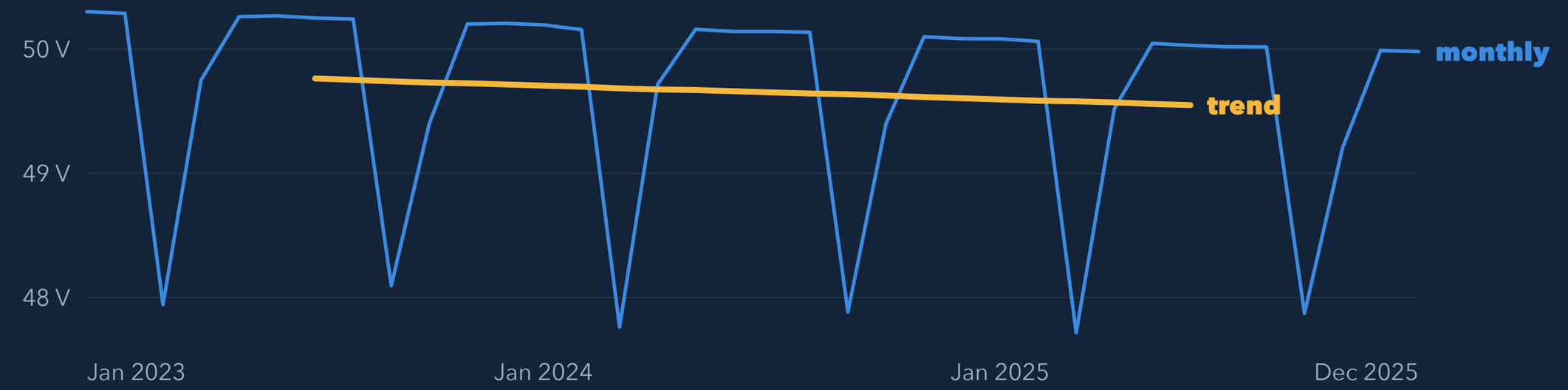


🌧️ Ka-band monthly: **36 months = 3 rainy seasons**. That's the minimum, not a comfortable margin. Say so in your report.

🗣️ One friend says the new buka is sweet: **gist**. Three different friends: **pattern**. Now you go.

BATTERY_HEALTH_DAILY (WEEK 3 VERSION) · END-OF-DISCHARGE VOLTAGE, MONTHLY

Not every season is one hump a year 🔋



🔁 Seasonal dips in **March (-1.89 V)** and **September (-1.70 V)**: two eclipse seasons, one at each equinox.

📉 Trend: **49.76 V → 49.55 V**. The battery is slowly ageing. Decomposition separates the two effects.

YOUR TURN · 60 MINUTES

Open the W2D1 exercise 👩🏾‍💻

1. Open in Colab → **File** → **Save a copy in Drive** 💾
2. **Part A:** take the Ka-band series apart
3. **Part B:** right period, right model, read the leftovers
4. Replace every ____, run, get the ✅



13 checks to pass

12 tasks + 🌶️ the robust-August bonus

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know

🔧 **observed = trend + seasonal + residual.** Multiply when the swing grows.

🥁 **The period comes from the process:** 24 · 7 · 12. Trust 3+ cycles.

🔍 **Investigate the residual.** `STL(robust=True)` for messy seasons.

Next time 👉 **Does Today Remember Yesterday?** Find the period, don't guess it. 🔁